Ho spiegato i fattoriali

nuova nei coefficienti binomiali

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

$$\frac{n!}{k!\,(n-k)!} = \frac{n!}{(n-k)!\,(n-k+k)!} = \frac{n!}{k!\,(n-k)!}$$

$$\binom{n}{k} \Rightarrow \binom{n-1}{k-1} + \binom{n-1}{k}$$

$$\frac{(n-1)!}{(k-1)!\,(n-k-1+1)!} + \frac{(n-1)!}{k!\,(n-1-k)!}$$

Abbiamo fatto il comune

$$\frac{(n-1)!}{(k-1)!\,(n-k)!} + \frac{(n-1)!}{k!\,(n-k-1)!} \rightarrow \frac{(n-1)!}{(k-1)!\,(n-k-1)!} \cdot \left(\frac{1}{n-k} + \frac{1}{k}\right)$$

$k \cdot (k-1)!$

$$\frac{(n-1)!}{(k-1)!\,(n-k-1)!} \cdot \frac{n}{k\,(n-k)}$$

$$\frac{n!}{(n-k)!\,k!}$$

$\bar{A}CD + B\bar{D} + \bar{A}\bar{C} + AB + AC + D\bar{A}\bar{B}$